

Newman-Penrose Formalism Calculations

Stephani et al., Equation (kerr-newman anti de sitter)

Metric

$$ds^2 = \left(\frac{\Delta_\theta a^2 \sin(\theta)^2 - \Delta_r}{\Xi^2 a^2 \cos(\theta)^2 + \Xi^2 r^2} \right) d\bar{t} \otimes d\bar{t} + \left(-\frac{(\Delta_\theta a^3 + \Delta_\theta a r^2 - \Delta_r a) \sin(\theta)^2}{\Xi^2 a^2 \cos(\theta)^2 + \Xi^2 r^2} \right) d\bar{t} \otimes d\bar{\phi} + \left(\frac{a^2 \cos(\theta)^2 + r^2}{\Delta_r} \right) dr \otimes dr + \left(\frac{a^2 \cos(\theta)^2 + r^2}{\Delta_\theta} \right) d\theta \otimes d\theta + \left(-\frac{(\Delta_\theta a^3 + \Delta_\theta a r^2 - \Delta_r a) \sin(\theta)^2}{\Xi^2 a^2 \cos(\theta)^2 + \Xi^2 r^2} \right) d\bar{\phi} \otimes d\bar{t} + \left(-\frac{\Delta_r a^2 \sin(\theta)^4 - (\Delta_\theta a^4 + 2 \Delta_\theta a^2 r^2 + \Delta_\theta r^4) \sin(\theta)^2}{\Xi^2 a^2 \cos(\theta)^2 + \Xi^2 r^2} \right) d\bar{\phi} \otimes d\bar{\phi}$$

Coordinates

$$(x^\mu) = (\bar{t}, r, \theta, \bar{\phi})$$

Null Tetrad

Covariant components

$$l = \left(-\frac{1}{\Xi} \right) d\bar{t} + \left(\frac{a^2 \cos(\theta)^2}{\Delta_r} + \frac{r^2}{\Delta_r} \right) dr + \left(\frac{a \sin(\theta)^2}{\Xi} \right) d\bar{\phi}$$

$$n = \left(-\frac{\Delta_r}{2(\Xi a^2 \cos(\theta)^2 + \Xi r^2)} \right) d\bar{t} + \left(-\frac{1}{2} \right) dr + \left(\frac{\Delta_r a \sin(\theta)^2}{2(\Xi a^2 \cos(\theta)^2 + \Xi r^2)} \right) d\bar{\phi}$$

$$m = \left(-\frac{\sqrt{2}\sqrt{\Delta_\theta} a^2 \cos(\theta) \sin(\theta)}{2(\Xi a^2 \cos(\theta)^2 + \Xi r^2)} - \frac{i \sqrt{2}\sqrt{\Delta_\theta} a r \sin(\theta)}{2(\Xi a^2 \cos(\theta)^2 + \Xi r^2)} \right) d\bar{t} + \left(-\frac{i \sqrt{2} a \cos(\theta)}{2\sqrt{\Delta_\theta}} + \frac{\sqrt{2} r}{2\sqrt{\Delta_\theta}} \right) d\theta + \left(\frac{\sqrt{2}\sqrt{\Delta_\theta} a^3 \cos(\theta) \sin(\theta)}{2(\Xi a^2 \cos(\theta)^2 + \Xi r^2)} + \frac{\sqrt{2}\sqrt{\Delta_\theta} a r^2 \cos(\theta) \sin(\theta)}{2(\Xi a^2 \cos(\theta)^2 + \Xi r^2)} + \frac{i \sqrt{2}\sqrt{\Delta_\theta} a^2 r \sin(\theta)}{2(\Xi a^2 \cos(\theta)^2 + \Xi r^2)} + \frac{i \sqrt{2}\sqrt{\Delta_\theta} r^3 \sin(\theta)}{2(\Xi a^2 \cos(\theta)^2 + \Xi r^2)} \right) d\bar{\phi}$$

$$\begin{aligned}\bar{m} = & \left(-\frac{\sqrt{2}\sqrt{\Delta_\theta}a^2 \cos(\theta) \sin(\theta)}{2(\Xi a^2 \cos(\theta)^2 + \Xi r^2)} + \frac{i\sqrt{2}\sqrt{\Delta_\theta}ar \sin(\theta)}{2(\Xi a^2 \cos(\theta)^2 + \Xi r^2)} \right) d\bar{t} + \left(\frac{i\sqrt{2}a \cos(\theta)}{2\sqrt{\Delta_\theta}} + \frac{\sqrt{2}r}{2\sqrt{\Delta_\theta}} \right) d\theta \\ & + \left(\frac{\sqrt{2}\sqrt{\Delta_\theta}a^3 \cos(\theta) \sin(\theta)}{2(\Xi a^2 \cos(\theta)^2 + \Xi r^2)} + \frac{\sqrt{2}\sqrt{\Delta_\theta}ar^2 \cos(\theta) \sin(\theta)}{2(\Xi a^2 \cos(\theta)^2 + \Xi r^2)} - \frac{i\sqrt{2}\sqrt{\Delta_\theta}a^2r \sin(\theta)}{2(\Xi a^2 \cos(\theta)^2 + \Xi r^2)} - \frac{i\sqrt{2}\sqrt{\Delta_\theta}r^3 \sin(\theta)}{2(\Xi a^2 \cos(\theta)^2 + \Xi r^2)} \right) d\bar{\phi}\end{aligned}$$

Contravariant components

$$l = \left(\frac{\Xi a^2}{\Delta_r} + \frac{\Xi r^2}{\Delta_r} \right) d\bar{t} + (1) dr + \left(\frac{\Xi a}{\Delta_r} \right) d\bar{\phi}$$

$$n = \left(\frac{\Xi a^2}{2(a^2 \cos(\theta)^2 + r^2)} + \frac{\Xi r^2}{2(a^2 \cos(\theta)^2 + r^2)} \right) d\bar{t} + \left(-\frac{\Delta_r}{2(a^2 \cos(\theta)^2 + r^2)} \right) dr + \left(\frac{\Xi a}{2(a^2 \cos(\theta)^2 + r^2)} \right) d\bar{\phi}$$

$$m = \left(\frac{i\sqrt{2}\Xi a \sin(\theta)}{2(i a \cos(\theta) + r)\sqrt{\Delta_\theta}} \right) d\bar{t} + \left(\frac{\sqrt{2}\sqrt{\Delta_\theta}}{2(i a \cos(\theta) + r)} \right) d\theta + \left(\frac{i\sqrt{2}\Xi}{2(i a \cos(\theta) + r)\sqrt{\Delta_\theta} \sin(\theta)} \right) d\bar{\phi}$$

$$\bar{m} = \left(-\frac{i\sqrt{2}\Xi a \sin(\theta)}{2(-i a \cos(\theta) + r)\sqrt{\Delta_\theta}} \right) d\bar{t} + \left(\frac{\sqrt{2}\sqrt{\Delta_\theta}}{2(-i a \cos(\theta) + r)} \right) d\theta + \left(-\frac{i\sqrt{2}\Xi}{2(-i a \cos(\theta) + r)\sqrt{\Delta_\theta} \sin(\theta)} \right) d\bar{\phi}$$

Directional Derivatives

$$D = \frac{\Xi a^2}{\Delta_r} + \frac{\Xi r^2}{\Delta_r} \partial_{\bar{t}} + \partial_r + \frac{\Xi a}{\Delta_r} \partial_{\bar{\phi}}$$

$$\Delta = \frac{\Xi a^2}{2(a^2 \cos(\theta)^2 + r^2)} + \frac{\Xi r^2}{2(a^2 \cos(\theta)^2 + r^2)} \partial_{\bar{t}} - \frac{\Delta_r}{2(a^2 \cos(\theta)^2 + r^2)} \partial_r + \frac{\Xi a}{2(a^2 \cos(\theta)^2 + r^2)} \partial_{\bar{\phi}}$$

$$\delta = \frac{i\sqrt{2}\Xi a \sin(\theta)}{2(i a \cos(\theta) + r)\sqrt{\Delta_\theta}} \partial_{\bar{t}} + \frac{\sqrt{2}\sqrt{\Delta_\theta}}{2(i a \cos(\theta) + r)} \partial_\theta + \frac{i\sqrt{2}\Xi}{2(i a \cos(\theta) + r)\sqrt{\Delta_\theta} \sin(\theta)} \partial_{\bar{\phi}}$$

$$\bar{\delta} = -\frac{i\sqrt{2}\Xi a \sin(\theta)}{2(-i a \cos(\theta) + r)\sqrt{\Delta_\theta}} \partial_{\bar{t}} + \frac{\sqrt{2}\sqrt{\Delta_\theta}}{2(-i a \cos(\theta) + r)} \partial_\theta - \frac{i\sqrt{2}\Xi}{2(-i a \cos(\theta) + r)\sqrt{\Delta_\theta} \sin(\theta)} \partial_{\bar{\phi}}$$

Spin Coefficients

$$\rho = -\frac{i a \cos(\theta)}{a^2 \cos(\theta)^2 + r^2} - \frac{r}{a^2 \cos(\theta)^2 + r^2}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{i \Delta_r a \cos(\theta)}{2 \left(a^4 \cos(\theta)^4 + 2 a^2 r^2 \cos(\theta)^2 + r^4 \right)} - \frac{\Delta_r r}{2 \left(a^4 \cos(\theta)^4 + 2 a^2 r^2 \cos(\theta)^2 + r^4 \right)}$$

$$\lambda = 0$$

$$\nu = 0$$

$\alpha =$

$$\begin{aligned}
& - \frac{\sqrt{\Delta_\theta} a^8 \cos(\theta)^9}{2 \left(i \sqrt{2} a^9 \cos(\theta)^9 - 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 + 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 - 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 + 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
& - \frac{3 i \sqrt{\Delta_\theta} a^7 r \cos(\theta)^8}{\left(i \sqrt{2} a^9 \cos(\theta)^9 - 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 + 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 - 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 + 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
& + \frac{\sqrt{\Delta_\theta} a^8 \cos(\theta)^7}{\left(i \sqrt{2} a^9 \cos(\theta)^9 - 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 + 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 - 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 + 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
& + \frac{7 \sqrt{\Delta_\theta} a^6 r^2 \cos(\theta)^7}{\left(i \sqrt{2} a^9 \cos(\theta)^9 - 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 + 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 - 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 + 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
& + \frac{7 i \sqrt{\Delta_\theta} a^7 r \cos(\theta)^6}{\left(i \sqrt{2} a^9 \cos(\theta)^9 - 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 + 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 - 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 + 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
& + \frac{7 i \sqrt{\Delta_\theta} a^5 r^3 \cos(\theta)^6}{\left(i \sqrt{2} a^9 \cos(\theta)^9 - 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 + 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 - 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 + 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
& - \frac{21 \sqrt{\Delta_\theta} a^6 r^2 \cos(\theta)^5}{\left(i \sqrt{2} a^9 \cos(\theta)^9 - 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 + 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 - 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 + 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
& - \dots
\end{aligned}$$

$$\begin{aligned}
\beta = & - \frac{i \sqrt{2} \sqrt{\Delta_\theta} a^5 \cos(\theta)^6}{4 \left(a^6 \cos(\theta)^6 - 6 i a^5 r \cos(\theta)^5 - 15 a^4 r^2 \cos(\theta)^4 + 20 i a^3 r^3 \cos(\theta)^3 + 15 a^2 r^4 \cos(\theta)^2 - 6 i a r^5 \cos(\theta) - r^6 \right) \sin(\theta)} \\
& - \frac{5 \sqrt{2} \sqrt{\Delta_\theta} a^4 r \cos(\theta)^5}{4 \left(a^6 \cos(\theta)^6 - 6 i a^5 r \cos(\theta)^5 - 15 a^4 r^2 \cos(\theta)^4 + 20 i a^3 r^3 \cos(\theta)^3 + 15 a^2 r^4 \cos(\theta)^2 - 6 i a r^5 \cos(\theta) - r^6 \right) \sin(\theta)} \\
& + \frac{5 i \sqrt{2} \sqrt{\Delta_\theta} a^3 r^2 \cos(\theta)^4}{2 \left(a^6 \cos(\theta)^6 - 6 i a^5 r \cos(\theta)^5 - 15 a^4 r^2 \cos(\theta)^4 + 20 i a^3 r^3 \cos(\theta)^3 + 15 a^2 r^4 \cos(\theta)^2 - 6 i a r^5 \cos(\theta) - r^6 \right) \sin(\theta)} \\
& + \frac{5 \sqrt{2} \sqrt{\Delta_\theta} a^2 r^3 \cos(\theta)^3}{2 \left(a^6 \cos(\theta)^6 - 6 i a^5 r \cos(\theta)^5 - 15 a^4 r^2 \cos(\theta)^4 + 20 i a^3 r^3 \cos(\theta)^3 + 15 a^2 r^4 \cos(\theta)^2 - 6 i a r^5 \cos(\theta) - r^6 \right) \sin(\theta)} \\
& - \frac{5 i \sqrt{2} \sqrt{\Delta_\theta} a r^4 \cos(\theta)^2}{4 \left(a^6 \cos(\theta)^6 - 6 i a^5 r \cos(\theta)^5 - 15 a^4 r^2 \cos(\theta)^4 + 20 i a^3 r^3 \cos(\theta)^3 + 15 a^2 r^4 \cos(\theta)^2 - 6 i a r^5 \cos(\theta) - r^6 \right) \sin(\theta)} \\
& - \frac{\sqrt{2} \sqrt{\Delta_\theta} r^5 \cos(\theta)}{4 \left(a^6 \cos(\theta)^6 - 6 i a^5 r \cos(\theta)^5 - 15 a^4 r^2 \cos(\theta)^4 + 20 i a^3 r^3 \cos(\theta)^3 + 15 a^2 r^4 \cos(\theta)^2 - 6 i a r^5 \cos(\theta) - r^6 \right) \sin(\theta)}
\end{aligned}$$

$$\begin{aligned}
\tau = & - \frac{i \sqrt{2} \sqrt{\Delta_\theta} a^9 \cos(\theta)^8 \sin(\theta)}{2 \left(a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10} \right)} \\
& - \frac{4 \sqrt{2} \sqrt{\Delta_\theta} a^8 r \cos(\theta)^7 \sin(\theta)}{a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10}} \\
& + \frac{14i \sqrt{2} \sqrt{\Delta_\theta} a^7 r^2 \cos(\theta)^6 \sin(\theta)}{a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10}} \\
& + \frac{28 \sqrt{2} \sqrt{\Delta_\theta} a^6 r^3 \cos(\theta)^5 \sin(\theta)}{a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10}} \\
& - \frac{35i \sqrt{2} \sqrt{\Delta_\theta} a^5 r^4 \cos(\theta)^4 \sin(\theta)}{a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10}} \\
& - \frac{28 \sqrt{2} \sqrt{\Delta_\theta} a^4 r^5 \cos(\theta)^3 \sin(\theta)}{a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10}} \\
& + \frac{14i \sqrt{2} \sqrt{\Delta_\theta} a^3 r^6 \cos(\theta)^2 \sin(\theta)}{a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10}} \\
& + \frac{4 \sqrt{2} \sqrt{\Delta_\theta} a^2 r^7 \cos(\theta) \sin(\theta)}{a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10}} \\
& - \frac{i \sqrt{2} \sqrt{\Delta_\theta} a r^8 \sin(\theta)}{2 \left(a^{10} \cos(\theta)^{10} - 8i a^9 r \cos(\theta)^9 - 27 a^8 r^2 \cos(\theta)^8 + 48i a^7 r^3 \cos(\theta)^7 + 42 a^6 r^4 \cos(\theta)^6 + 42 a^4 r^6 \cos(\theta)^4 - 48i a^3 r^7 \cos(\theta)^3 - 27 a^2 r^8 \cos(\theta)^2 + 8i a r^9 \cos(\theta) + r^{10} \right)}
\end{aligned}$$

$$\begin{aligned}
\pi = & - \frac{i \sqrt{2} \sqrt{\Delta_\theta} a^7 \cos(\theta)^6 \sin(\theta)}{2 \left(a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8 \right)} \\
& + \frac{3 \sqrt{2} \sqrt{\Delta_\theta} a^6 r \cos(\theta)^5 \sin(\theta)}{a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8} \\
& + \frac{15i \sqrt{2} \sqrt{\Delta_\theta} a^5 r^2 \cos(\theta)^4 \sin(\theta)}{2 \left(a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8 \right)} \\
& - \frac{10 \sqrt{2} \sqrt{\Delta_\theta} a^4 r^3 \cos(\theta)^3 \sin(\theta)}{a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8} \\
& - \frac{15i \sqrt{2} \sqrt{\Delta_\theta} a^3 r^4 \cos(\theta)^2 \sin(\theta)}{2 \left(a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8 \right)} \\
& + \frac{3 \sqrt{2} \sqrt{\Delta_\theta} a^2 r^5 \cos(\theta) \sin(\theta)}{a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8} \\
& + \frac{i \sqrt{2} \sqrt{\Delta_\theta} a r^6 \sin(\theta)}{2 \left(a^8 \cos(\theta)^8 + 8i a^7 r \cos(\theta)^7 - 28 a^6 r^2 \cos(\theta)^6 - 56i a^5 r^3 \cos(\theta)^5 + 70 a^4 r^4 \cos(\theta)^4 + 56i a^3 r^5 \cos(\theta)^3 - 28 a^2 r^6 \cos(\theta)^2 - 8i a r^7 \cos(\theta) + r^8 \right)}
\end{aligned}$$

$$\epsilon = 0$$

$$\gamma$$

$$\begin{aligned}
= & \frac{\Delta_r a^{10} \cos(\theta)^{10}}{2 \left(i a^{13} \cos(\theta)^{13} + 9 a^{12} r \cos(\theta)^{12} - 34i a^{11} r^2 \cos(\theta)^{11} - 66 a^{10} r^3 \cos(\theta)^{10} + 55i a^9 r^4 \cos(\theta)^9 - 33 a^8 r^5 \cos(\theta)^8 + 132i a^7 r^6 \cos(\theta)^7 + 132 a^6 r^7 \cos(\theta)^6 - 33i a^5 r^8 \cos(\theta)^5 + 55 a^4 r^9 \cos(\theta)^4 - 33i a^3 r^{10} \cos(\theta)^3 + 132 a^2 r^{11} \cos(\theta)^2 - 132 i a r^{12} \cos(\theta) + r^{13} \right)} \\
& - \frac{5i \Delta_r a^9 r \cos(\theta)^9}{i a^{13} \cos(\theta)^{13} + 9 a^{12} r \cos(\theta)^{12} - 34i a^{11} r^2 \cos(\theta)^{11} - 66 a^{10} r^3 \cos(\theta)^{10} + 55i a^9 r^4 \cos(\theta)^9 - 33 a^8 r^5 \cos(\theta)^8 + 132i a^7 r^6 \cos(\theta)^7 + 132 a^6 r^7 \cos(\theta)^6 - 33i a^5 r^8 \cos(\theta)^5 + 55 a^4 r^9 \cos(\theta)^4 - 33i a^3 r^{10} \cos(\theta)^3 + 132 a^2 r^{11} \cos(\theta)^2 - 132 i a r^{12} \cos(\theta) + r^{13}} \\
& - \frac{45 \Delta_r a^8 r^2 \cos(\theta)^8}{2 \left(i a^{13} \cos(\theta)^{13} + 9 a^{12} r \cos(\theta)^{12} - 34i a^{11} r^2 \cos(\theta)^{11} - 66 a^{10} r^3 \cos(\theta)^{10} + 55i a^9 r^4 \cos(\theta)^9 - 33 a^8 r^5 \cos(\theta)^8 + 132i a^7 r^6 \cos(\theta)^7 + 132 a^6 r^7 \cos(\theta)^6 - 33i a^5 r^8 \cos(\theta)^5 + 55 a^4 r^9 \cos(\theta)^4 - 33i a^3 r^{10} \cos(\theta)^3 + 132 a^2 r^{11} \cos(\theta)^2 - 132 i a r^{12} \cos(\theta) + r^{13} \right)} \\
& + \frac{60i \Delta_r a^7 r^3 \cos(\theta)^7}{i a^{13} \cos(\theta)^{13} + 9 a^{12} r \cos(\theta)^{12} - 34i a^{11} r^2 \cos(\theta)^{11} - 66 a^{10} r^3 \cos(\theta)^{10} + 55i a^9 r^4 \cos(\theta)^9 - 33 a^8 r^5 \cos(\theta)^8 + 132i a^7 r^6 \cos(\theta)^7 + 132 a^6 r^7 \cos(\theta)^6 - 33i a^5 r^8 \cos(\theta)^5 + 55 a^4 r^9 \cos(\theta)^4 - 33i a^3 r^{10} \cos(\theta)^3 + 132 a^2 r^{11} \cos(\theta)^2 - 132 i a r^{12} \cos(\theta) + r^{13}} \\
& + \frac{105 \Delta_r a^6 r^4 \cos(\theta)^6}{i a^{13} \cos(\theta)^{13} + 9 a^{12} r \cos(\theta)^{12} - 34i a^{11} r^2 \cos(\theta)^{11} - 66 a^{10} r^3 \cos(\theta)^{10} + 55i a^9 r^4 \cos(\theta)^9 - 33 a^8 r^5 \cos(\theta)^8 + 132i a^7 r^6 \cos(\theta)^7 + 132 a^6 r^7 \cos(\theta)^6 - 33i a^5 r^8 \cos(\theta)^5 + 55 a^4 r^9 \cos(\theta)^4 - 33i a^3 r^{10} \cos(\theta)^3 + 132 a^2 r^{11} \cos(\theta)^2 - 132 i a r^{12} \cos(\theta) + r^{13}} \\
& - \frac{126i \Delta_r a^5 r^5 \cos(\theta)^5}{i a^{13} \cos(\theta)^{13} + 9 a^{12} r \cos(\theta)^{12} - 34i a^{11} r^2 \cos(\theta)^{11} - 66 a^{10} r^3 \cos(\theta)^{10} + 55i a^9 r^4 \cos(\theta)^9 - 33 a^8 r^5 \cos(\theta)^8 + 132i a^7 r^6 \cos(\theta)^7 + 132 a^6 r^7 \cos(\theta)^6 - 33i a^5 r^8 \cos(\theta)^5 + 55 a^4 r^9 \cos(\theta)^4 - 33i a^3 r^{10} \cos(\theta)^3 + 132 a^2 r^{11} \cos(\theta)^2 - 132 i a r^{12} \cos(\theta) + r^{13}} \\
& - \dots
\end{aligned}$$

Relation between Spin Coefficients

$$\begin{aligned}
\bar{\tau} + \pi = & \frac{i\sqrt{2}\sqrt{\Delta_\theta}a^9\cos(\theta)^8\sin(\theta)}{2\left(a^{10}\cos(\theta)^{10}+8ia^9r\cos(\theta)^9-27a^8r^2\cos(\theta)^8-48ia^7r^3\cos(\theta)^7+42a^6r^4\cos(\theta)^6+42a^4r^6\cos(\theta)^4+48ia^3r^7\cos(\theta)^3-27a^2r^8\cos(\theta)^2-8iar^9\cos(\theta)+r^{10}\right)} \\
& - \frac{4\sqrt{2}\sqrt{\Delta_\theta}a^8r\cos(\theta)^7\sin(\theta)}{a^{10}\cos(\theta)^{10}+8ia^9r\cos(\theta)^9-27a^8r^2\cos(\theta)^8-48ia^7r^3\cos(\theta)^7+42a^6r^4\cos(\theta)^6+42a^4r^6\cos(\theta)^4+48ia^3r^7\cos(\theta)^3-27a^2r^8\cos(\theta)^2-8iar^9\cos(\theta)+r^{10}} \\
& - \frac{14i\sqrt{2}\sqrt{\Delta_\theta}a^7r^2\cos(\theta)^6\sin(\theta)}{a^{10}\cos(\theta)^{10}+8ia^9r\cos(\theta)^9-27a^8r^2\cos(\theta)^8-48ia^7r^3\cos(\theta)^7+42a^6r^4\cos(\theta)^6+42a^4r^6\cos(\theta)^4+48ia^3r^7\cos(\theta)^3-27a^2r^8\cos(\theta)^2-8iar^9\cos(\theta)+r^{10}} \\
& + \frac{28\sqrt{2}\sqrt{\Delta_\theta}a^6r^3\cos(\theta)^5\sin(\theta)}{a^{10}\cos(\theta)^{10}+8ia^9r\cos(\theta)^9-27a^8r^2\cos(\theta)^8-48ia^7r^3\cos(\theta)^7+42a^6r^4\cos(\theta)^6+42a^4r^6\cos(\theta)^4+48ia^3r^7\cos(\theta)^3-27a^2r^8\cos(\theta)^2-8iar^9\cos(\theta)+r^{10}} \\
& + \frac{35i\sqrt{2}\sqrt{\Delta_\theta}a^5r^4\cos(\theta)^4\sin(\theta)}{a^{10}\cos(\theta)^{10}+8ia^9r\cos(\theta)^9-27a^8r^2\cos(\theta)^8-48ia^7r^3\cos(\theta)^7+42a^6r^4\cos(\theta)^6+42a^4r^6\cos(\theta)^4+48ia^3r^7\cos(\theta)^3-27a^2r^8\cos(\theta)^2-8iar^9\cos(\theta)+r^{10}} \\
& - \frac{i\sqrt{2}\sqrt{\Delta_\theta}a^7\cos(\theta)^6\sin(\theta)}{2\left(a^8\cos(\theta)^8+8ia^7r\cos(\theta)^7-28a^6r^2\cos(\theta)^6-56ia^5r^3\cos(\theta)^5+70a^4r^4\cos(\theta)^4+56ia^3r^5\cos(\theta)^3-28a^2r^6\cos(\theta)^2-8iar^7\cos(\theta)+r^8\right)} \\
& - \frac{28\sqrt{2}\sqrt{\Delta_\theta}a^4r^5\cos(\theta)^3\sin(\theta)}{a^{10}\cos(\theta)^{10}+8ia^9r\cos(\theta)^9-27a^8r^2\cos(\theta)^8-48ia^7r^3\cos(\theta)^7+42a^6r^4\cos(\theta)^6+42a^4r^6\cos(\theta)^4+48ia^3r^7\cos(\theta)^3-27a^2r^8\cos(\theta)^2-8iar^9\cos(\theta)+r^{10}} \\
& + \frac{3\sqrt{2}\sqrt{\Delta_\theta}a^6r\cos(\theta)^5\sin(\theta)}{a^8\cos(\theta)^8+8ia^7r\cos(\theta)^7-28a^6r^2\cos(\theta)^6-56ia^5r^3\cos(\theta)^5+70a^4r^4\cos(\theta)^4+56ia^3r^5\cos(\theta)^3-28a^2r^6\cos(\theta)^2-8iar^7\cos(\theta)+r^8} \\
& - \frac{14i\sqrt{2}\sqrt{\Delta_\theta}a^3r^6\cos(\theta)^2\sin(\theta)}{a^{10}\cos(\theta)^{10}+8ia^9r\cos(\theta)^9-27a^8r^2\cos(\theta)^8-48ia^7r^3\cos(\theta)^7+42a^6r^4\cos(\theta)^6+42a^4r^6\cos(\theta)^4+48ia^3r^7\cos(\theta)^3-27a^2r^8\cos(\theta)^2-8iar^9\cos(\theta)+r^{10}} + \dots
\end{aligned}$$

$$\bar{\alpha} + \beta$$

$$\begin{aligned}
&= \frac{\sqrt{\Delta_\theta} a^8 \cos(\theta)^9}{2 \left(i \sqrt{2} a^9 \cos(\theta)^9 + 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 - 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 + 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 - 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
&\quad - \frac{3 i \sqrt{\Delta_\theta} a^7 r \cos(\theta)^8}{\left(i \sqrt{2} a^9 \cos(\theta)^9 + 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 - 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 + 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 - 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
&\quad - \frac{\sqrt{\Delta_\theta} a^8 \cos(\theta)^7}{\left(i \sqrt{2} a^9 \cos(\theta)^9 + 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 - 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 + 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 - 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
&\quad - \frac{7 \sqrt{\Delta_\theta} a^6 r^2 \cos(\theta)^7}{\left(i \sqrt{2} a^9 \cos(\theta)^9 + 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 - 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 + 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 - 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
&\quad + \frac{7 i \sqrt{\Delta_\theta} a^7 r \cos(\theta)^6}{\left(i \sqrt{2} a^9 \cos(\theta)^9 + 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 - 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 + 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 - 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
&\quad + \frac{7 i \sqrt{\Delta_\theta} a^5 r^3 \cos(\theta)^6}{\left(i \sqrt{2} a^9 \cos(\theta)^9 + 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 - 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 + 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 - 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
&\quad + \frac{21 \sqrt{\Delta_\theta} a^6 r^2 \cos(\theta)^5}{\left(i \sqrt{2} a^9 \cos(\theta)^9 + 9 \sqrt{2} a^8 r \cos(\theta)^8 - 36 i \sqrt{2} a^7 r^2 \cos(\theta)^7 - 84 \sqrt{2} a^6 r^3 \cos(\theta)^6 + 126 i \sqrt{2} a^5 r^4 \cos(\theta)^5 + 126 \sqrt{2} a^4 r^5 \cos(\theta)^4 - 84 i \sqrt{2} a^3 r^6 \cos(\theta)^3 - 36 \sqrt{2} a^2 r^7 \cos(\theta)^2 + 9 i \sqrt{2} a r^8 \cos(\theta) - r^9 \right)} \\
&\quad - \dots
\end{aligned}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{\Delta_\theta a^2 \sin(\theta)^2}{4 \left(a^4 \cos(\theta)^4 + 2 a^2 r^2 \cos(\theta)^2 + r^4 \right)} - \frac{3 \Delta_\theta a^2}{4 \left(a^4 \cos(\theta)^4 + 2 a^2 r^2 \cos(\theta)^2 + r^4 \right)} + \frac{\Delta_\theta r^2}{4 \left(a^4 \cos(\theta)^4 + 2 a^2 r^2 \cos(\theta)^2 + r^4 \right)} + \frac{\Delta_r}{2 \left(a^4 \cos(\theta)^4 + 2 a^2 r^2 \cos(\theta)^2 + r^4 \right)}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = \frac{\Delta_\theta}{12 \left(a^2 \cos(\theta)^2 + r^2 \right)}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\begin{aligned} \Psi_2 = & -\frac{\Delta_\theta a^4 \sin(\theta)^4}{6 \left(a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6 \right)} - \frac{i \Delta_\theta a^3 r \cos(\theta) \sin(\theta)^2}{a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6} - \frac{2 \Delta_\theta a^4 \sin(\theta)^2}{3 \left(a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6 \right)} \\ & - \frac{5 \Delta_\theta a^2 r^2 \sin(\theta)^2}{3 \left(a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6 \right)} - \frac{i \Delta_\theta a^3 r \cos(\theta)}{a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6} - \frac{i \Delta_\theta a r^3 \cos(\theta)}{a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6} \\ & + \frac{5 \Delta_\theta a^4}{6 \left(a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6 \right)} + \frac{2 \Delta_\theta a^2 r^2}{3 \left(a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6 \right)} \\ & - \frac{\Delta_\theta r^4}{6 \left(a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6 \right)} + \frac{\Delta_r a^2 \sin(\theta)^2}{a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6} \\ & + \frac{2i \Delta_r a r \cos(\theta)}{a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6} - \frac{\Delta_r a^2}{a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6} + \frac{\Delta_r r^2}{a^6 \cos(\theta)^6 + 3 a^4 r^2 \cos(\theta)^4 + 3 a^2 r^4 \cos(\theta)^2 + r^6} \end{aligned}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"